

TOBY LAU

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EDUCATION

University of British Columbia

B.Sc. Computer Science – GPA: 3.95

May 2028 (Expected)

Vancouver, BC

EXPERIENCE

Squarepoint Capital

Montreal, QC

Software Engineer Intern – Data Services (Portfolio Construction & Trading)

Sep 2026 – Dec 2026

- Building a Barra risk factor analytics platform in Python and q for PMs to decompose strategy holdings and trade baskets.
- Implemented a qunit-based testing framework for a kdb+/q web service, mocking live services with recorded responses.
- Consolidated ~10 microservices into 2 product monorepos, unifying CI/CD pipelines and containerizing dev environments.

Software Engineer Intern – Data Services (Commodities)

May 2026 – Aug 2026

- Created a monitoring and alerting platform to aggregate and visualize health check results across 1100+ time-series alphas.
- Cut p95 API latency by 22% by merging an Apollo subgraph with downstream gRPC services across 35+ prod endpoints.
- Built a distributed poller/worker data pipeline using Python ARQ to ingest, process, and store ~20k alpha metrics/second.
- Implemented cron-derived lookback window calculation and status detection to detect stale metrics across 100+ strategies.
- Designed a YAML-database sync mechanism enabling 50+ desks to version-control dashboard configurations through GitLab.

Planview

Vancouver, BC

Software Engineer Intern – Planview Copilot

Jan 2025 – Aug 2025

- Reduced chat response latency by 50%+ by implementing concurrent tool calls to AWS Lambdas in Java-based chat service.
- Unblocked a major product release for 2k+ users by implementing a dual-token handler during an auth service migration.
- Implemented a log obfuscator/forwarder that routed 21k+ sanitized daily logs to Datadog, retaining raw logs in CloudWatch.
- Delivered a self-service RAG platform for 100+ tenants with S3 and Bedrock, enabling document uploads and analysis/Q&A.
- Built a hybrid tool-calling/MCP chat service and registry to support existing tools while integrating newer MCP servers.

UBC Department of Computer Science

Vancouver, BC

Undergraduate Teaching Assistant – CPSC 210 (Introduction to Object Oriented Programming)

Sep 2024 – Dec 2024

- Taught OOP/OOD, design patterns, and TDD with JUnit to 300+ students; mentored 25+ students in weekly labs/office hours.

APL Zuellig Pharma

Remote

Software Engineer Intern – Distribution Resource Planning

Oct 2021 – Jan 2022

- Replaced a complex Excel workflow for stock/distribution planning with a Python GUI across 46k+ SKUs in 25 warehouses.

EXTRACURRICULARS

Databricks

Remote

Student Fellow

Jun 2026 – Present

- 1 of 36 selected from 5k+ globally to represent Databricks campus/user groups, with mentorship from Databricks engineers.

PROJECTS

Extreme Vocals Classifier | Python, PyTorch, Numpy, Scipy, Librosa, Google Colab

- Synthesized a 500+ extreme vocal sample dataset by isolating vocals with Demucs, extracting and processing relevant vocal regions with Numpy/Scipy (RMS normalization, high-pass filter), and augmenting vocals with librosa from 57+ songs.

Membership Portal | React, TypeScript, PostgreSQL, Stripe, Supabase, Google Cloud Platform

- Supported 250+ MAU and \$5k+ growth in revenue by building payments, registration flows, and admin tools from 0→1.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, HTML, CSS, SQL, C, C++, kdb/q+

Frameworks: React, Spring (Boot), Next.js, Express, Flask, FastAPI, JUnit, Jest

Technologies: AWS, GCP, Supabase, PostgreSQL, Docker, Kubernetes, ArgoCD, Liquibase/Alembic, GraphQL, gRPC

Tools: Git, Postman/Insomnia, Datadog, Elastic Search, Jupyter, DBeaver, Vim, Figma, Jira, IntelliJ, Claude Code